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MASTER 1 : Réseaux et Systèmes Informatique

Rapport Virtualisation de Datacenter avec Proxmox et Conteneurs

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Travaux Pratiques – Virtualisation de Datacenter avec Proxmox et Conteneurs

Ce document propose une série de TP axés sur l'usage de Proxmox VE, incluant la virtualisation de machines et l'utilisation de conteneurs (LXC).

Prérequis - Serveur ou machine virtuelle avec Proxmox VE installé - Accès à l'interface web Proxmox (port 8006) - Images ISO Linux (Ubuntu/CentOS) - Connexion réseau active - Conteneurs LXC disponibles depuis les templates Proxmox

TP1 Installation de Proxmox VE

** Installer Proxmox VE sur une machine physique ou une VM.

** Configurer le réseau, les dépôts APT, et accéder à l'interface Web via <https://IP:8006>.

```
-----

Welcome to the Proxmox Virtual Environment. Please use your web browser to
configure this server - connect to:

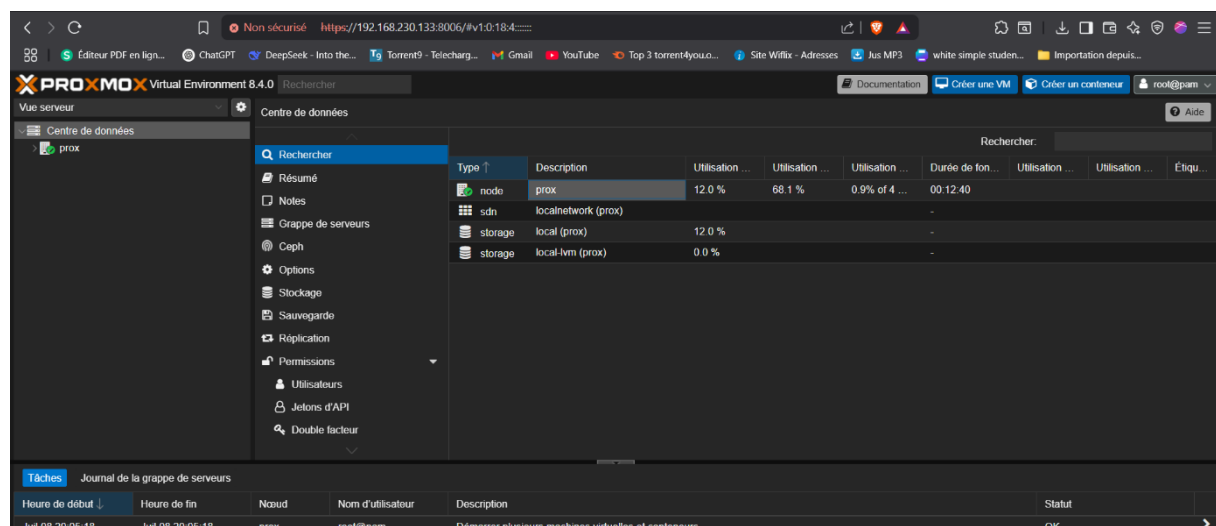
https://192.168.230.133:8006/

-----

prox login: root
Password:
Linux prox 6.8.12-9-pve #1 SMP PREEMPT_DYNAMIC PMX 6.8.12-9 (2025-03-16T19:18Z) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@prox:~# _
```



TP2 – Création d'une VM Linux

** Téléverser une image ISO (ex. Ubuntu Server) dans le stockage ISO et créer une VM avec 2 vCPU, 2048 Mo RAM, 20 Go disque, et installer Ubuntu.

Créer: Machine virtuelle

Général Système d'exploitation Système Disques Processeur Mémoire Réseau **Confirmation**

Key ↑	Value
cores	2
cpu	x86-64-v2-AES
ide2	local:iso/ubuntu-24.04.2-desktop-amd64.iso,media=cdrom
memory	2048
net0	virtio,bridge=vibr0,firewall=1
nodename	prox
numa	0
ostype	l26
scsi0	local-lvm:20,iothread=on
scsihw	virtio-scsi-single
sockets	2
vmid	100

☐ Démarrer après création

Avancé ☐ **Retour** **Terminer**

Voici le VM Ubuntu

Install

Who are you?

Your name: Ousmane ✓

Your computer's name: ousmane-Standard-PC ✓
The name it uses when it talks to other computers.

Pick a username: ousmane ✓

Choose a password: ● Short password

Confirm your password: ● ✓

☐ Log in automatically

☒ Require my password to log in

Back Continue

**** Configurer un accès SSH à la VM et tester la connectivité.**

```
ousmane@ousmane-Standard-PC-i440FX-PIIX-1996:~$ sudo su
[sudo] password for ousmane:
root@ousmane-Standard-PC-i440FX-PIIX-1996:/home/ousmane# sudo apt update && sudo
apt install openssh-server -y
Hit:1 http://sn.archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://sn.archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Hit:3 http://security.ubuntu.com/ubuntu focal-security InRelease
Get:4 http://sn.archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Get:5 http://sn.archive.ubuntu.com/ubuntu focal-updates/main i386 Packages [1114
kB]
Get:6 http://sn.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [395
4 kB]
46% [6 Packages 1405 kB/3954 kB 36%]
```

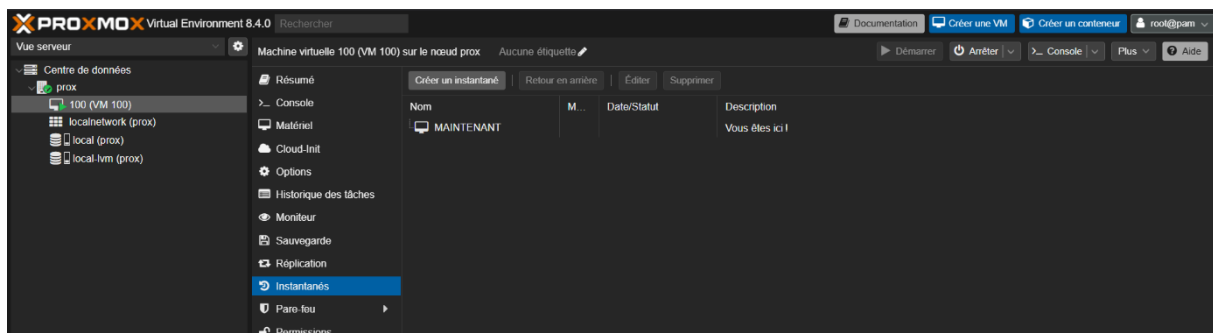
Le service tourne bien :

```
root@ousmane-Standard-PC-i440FX-PIIX-1996:/home/ousmane# sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2025-07-09 13:12:06 GMT; 6min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 4527 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 4529 (sshd)
       Tasks: 1 (limit: 2256)
      Memory: 1.1M
    CGroup: /system.slice/ssh.service
            └─4529 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups

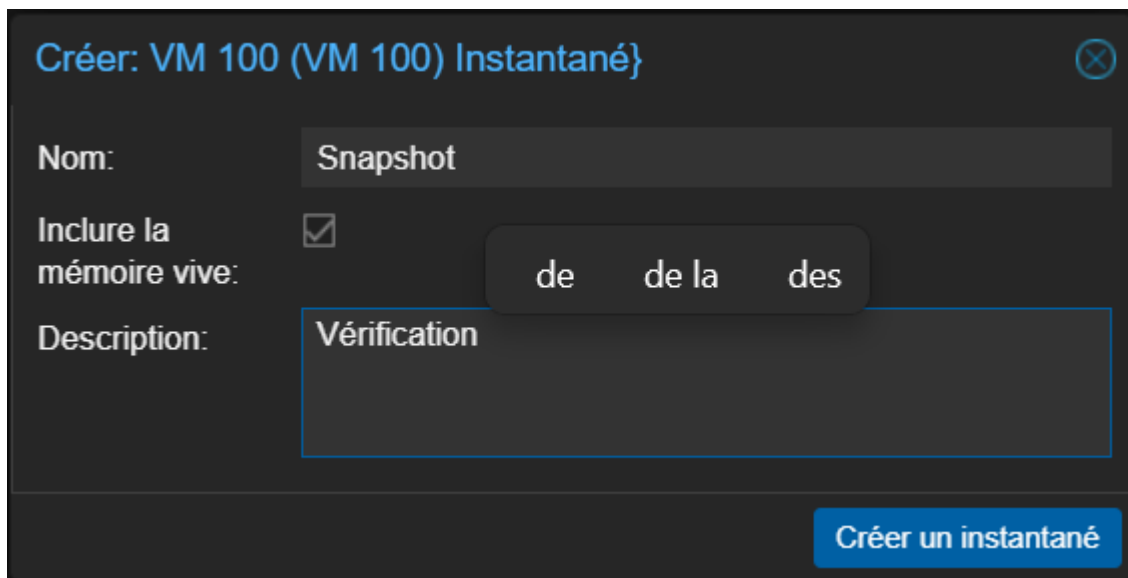
sul 09 13:12:06 ousmane-Standard-PC-i440FX-PIIX-1996 systemd[1]: Starting OpenBSD Secure Shell server...
sul 09 13:12:06 ousmane-Standard-PC-i440FX-PIIX-1996 sshd[4529]: Server listening on 0.0.0.0 port 22.
sul 09 13:12:06 ousmane-Standard-PC-i440FX-PIIX-1996 sshd[4529]: Server listening on :: port 22.
sul 09 13:12:06 ousmane-Standard-PC-i440FX-PIIX-1996 systemd[1]: Started OpenBSD Secure Shell server.
sul 09 13:16:45 ousmane-Standard-PC-i440FX-PIIX-1996 sshd[4538]: Connection reset by 192.168.230.1 port 61615 [preauth]
sul 09 13:18:15 ousmane-Standard-PC-i440FX-PIIX-1996 sshd[4544]: Connection reset by 192.168.230.1 port 61622 [preauth]
root@ousmane-Standard-PC-i440FX-PIIX-1996:/home/ousmane#
```

TP3 – Gestion des ressources et snapshots

****Créer un snapshot de la VM précédente.**



Donner un nom :



Créer: VM 100 (VM 100) Instantané

Nom: Snapshot

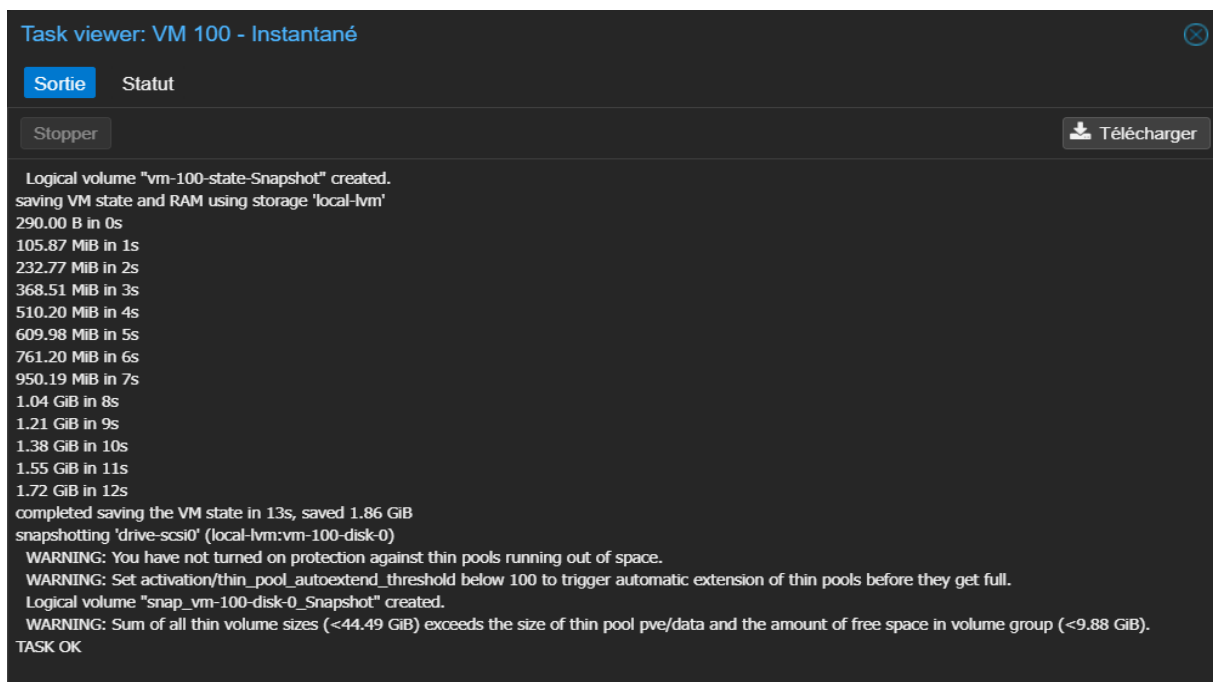
Inclure la mémoire vive: ☒

Description: Vérification

de de la des

Créer un instantané

Le snapshot est créé :



Task viewer: VM 100 - Instantané

Sortie Statut

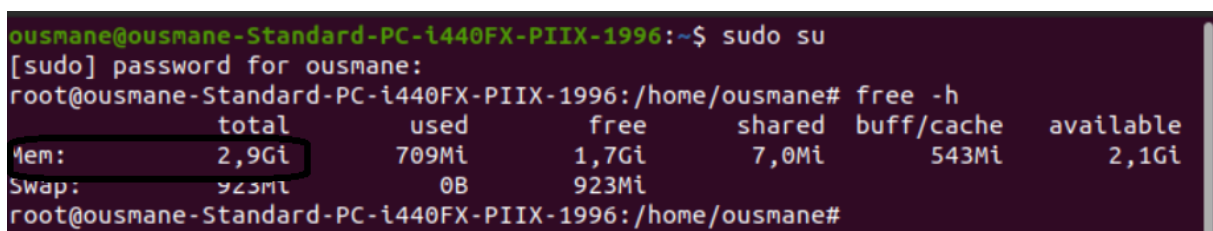
Stopper Télécharger

```
Logical volume "vm-100-state-Snapshot" created.
saving VM state and RAM using storage 'local-lvm'
290.00 B in 0s
105.87 MiB in 1s
232.77 MiB in 2s
368.51 MiB in 3s
510.20 MiB in 4s
609.98 MiB in 5s
761.20 MiB in 6s
950.19 MiB in 7s
1.04 GiB in 8s
1.21 GiB in 9s
1.38 GiB in 10s
1.55 GiB in 11s
1.72 GiB in 12s
completed saving the VM state in 13s, saved 1.86 GiB
snapshotting 'drive-scsi0' (local-lvm:vm-100-disk-0)
WARNING: You have not turned on protection against thin pools running out of space.
WARNING: Set activation/thin_pool_autoextend_threshold below 100 to trigger automatic extension of thin pools before they get full.
Logical volume "snap_vm-100-disk-0_Snapshot" created.
WARNING: Sum of all thin volume sizes (<44.49 GiB) exceeds the size of thin pool pve/data and the amount of free space in volume group (<9.88 GiB).
TASK OK
```

** Augmenter les ressources CPU/RAM et observer les effets.

Après augmentation les nouveaux paramètres sont pris en compte :

RAM :

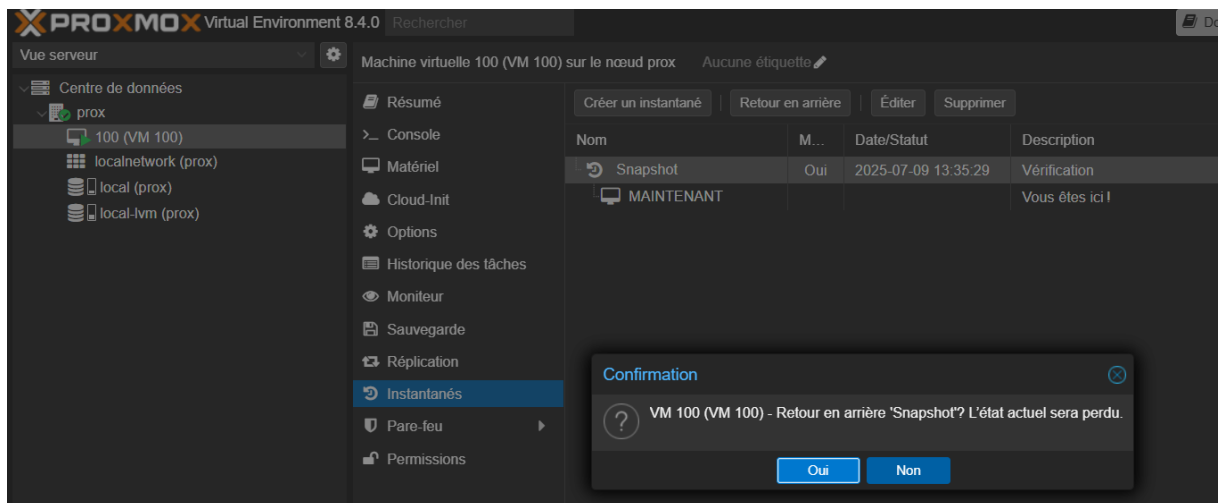


```
ousmane@ousmane-Standard-PC-i440FX-PIIX-1996:~$ sudo su
[sudo] password for ousmane:
root@ousmane-Standard-PC-i440FX-PIIX-1996:/home/ousmane# free -h
              total        used        free      shared  buff/cache   available
Mem:          2,9Gi         709Mi        1,7Gi         7,0Mi        543Mi        2,1Gi
Swap:          923Mi           0B         923Mi
```

CPU :

```
root@ousmane-Standard-PC-i440FX-PIIX-1996:/home/ousmane# lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Byte Order:             Little Endian
Address sizes:          40 bits physical, 48 bits virtual
CPU(s):                 3
```

****Restaurer le snapshot.**



Après avoir restaurer le snapshot on remarque que les paramètres qu'on avait changé sont ont repris les valeurs de base qu'on avait mis en créant la VM:

RAM :

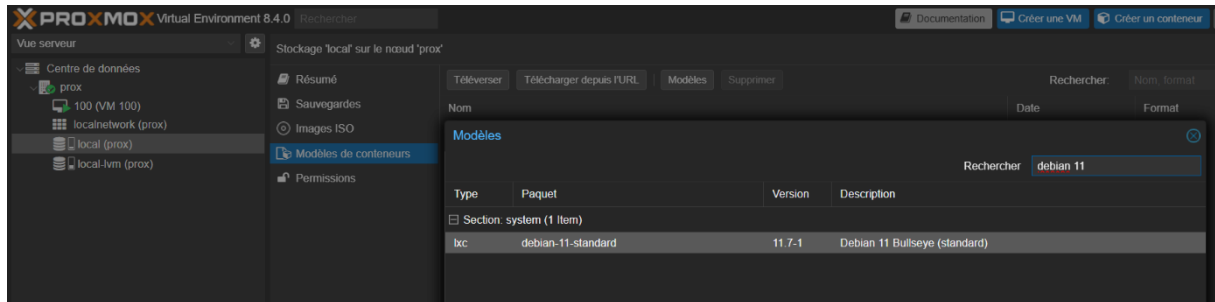
```
root@ousmane-Standard-PC-i440FX-PIIX-1996:/home/ousmane# free -h
               total        used        free      shared  buff/cache   available
Mem:           1,9Gi        669Mi        758Mi         7,0Mi         535Mi        1,1Gi
Swap:          923Mi           0B         923Mi
```

CPU :

```
root@ousmane-Standard-PC-i440FX-PIIX-1996:/home/ousmane# lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Byte Order:             Little Endian
Address sizes:          40 bits physical, 48 bits virtual
CPU(s):                 2
```

TP4 – Déploiement et gestion d'un conteneur LXC

****Télécharger un Template LXC depuis l'interface Proxmox (ex. Debian 11).**



Cliquez sur créer un conteneur, donner un nom et le mot de passe :

The screenshot shows the 'Créer: Conteneur LXC' form. It has tabs for 'Général', 'Modèle', 'Disques', 'Processeur', 'Mémoire', 'Réseau', 'DNS', and 'Confirmation'. The 'Général' tab is selected. The form contains the following fields and options:

- Nœud:** prox (dropdown)
- CT ID:** 101 (dropdown)
- Nom d'hôte:** debian11 (text input)
- Conteneur non privilégié:** ☒
- Imbriqué:** ☒
- Pool de ressources:** (dropdown)
- Mot de passe:** (password input)
- Confirmer le mot de passe:** (password input)
- Clef(s) SSH publique(s):** (text area)
- Charger le fichier de clef SSH** (button)

At the bottom, there is an 'Aide' button, an 'Avancé' checkbox, and 'Retour' and 'Suivant' buttons.

Choisir le conteneur préalablement téléchargé :

Créer: Conteneur LXC

Général **Modèle** Disques Processeur Mémoire Réseau DNS Confirmation

Stockage: local

Modèle: debian-11-standard_11.7-1_amd64

Créer un conteneur avec 1 vCPU, 512 Mo RAM, et accès au réseau bridge.

Créer: Conteneur LXC

Général Modèle Disques Processeur Mémoire Réseau DNS **Confirmation**

Key ↑	Value
cores	1
features	nesting=1
hostname	debian11
memory	512
net0	name=eth0,bridge=vmbr0,firewall=1
nodename	prox
ostemplate	local:vztmpl/debian-11-standard_11.7-1_amd64.tar.zst
pool	
rootfs	local-lvm:8
ssh-public-keys	
swap	512
unprivileged	1
vmid	101

☐ Démarrer après création

Avancé ☐ Retour Terminer

****Installer et lancer un service simple (Apache ou Nginx).**

Installation et démarrage de nginx :

```
root@debian11:~# apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  fontconfig-config fonts-dejavu-core geoip-database libdeflate0 libfontconfig1 libfreetype6 libgd3 libgeoip1
  libjbig0 libjpeg62-turbo libnginx-mod-http-geoip libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter
  libnginx-mod-mail libnginx-mod-stream libnginx-mod-stream-geoip libpng16-16 libtiff5 libwebp6 libx11-6
  libx11-data libxau6 libxcb1 libxdmcp6 libxpm4 libxslt1.1 nginx-common nginx-core
Suggested packages:
  libgd-tools geoip-bin fcgiwrap nginx-doc
The following NEW packages will be installed:
  fontconfig-config fonts-dejavu-core geoip-database libdeflate0 libfontconfig1 libfreetype6 libgd3 libgeoip1
  libjbig0 libjpeg62-turbo libnginx-mod-http-geoip libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter
  libnginx-mod-mail libnginx-mod-stream libnginx-mod-stream-geoip libpng16-16 libtiff5 libwebp6 libx11-6
  libx11-data libxau6 libxcb1 libxdmcp6 libxpm4 libxslt1.1 nginx-common nginx-core
```

Etat du serveur OK :

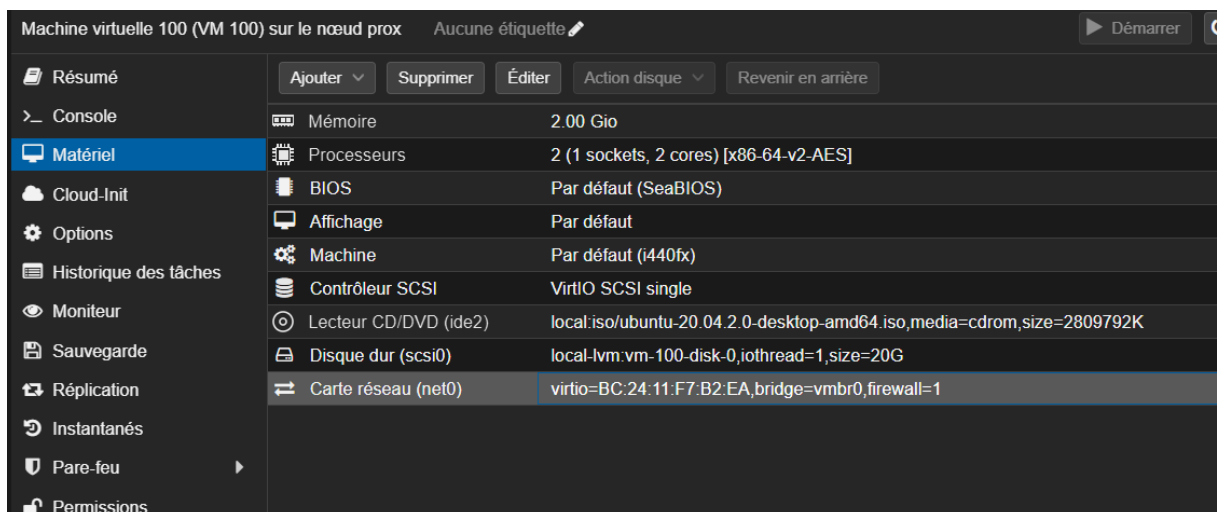
```
root@debian11:~# service nginx start
root@debian11:~# service nginx status
* nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2025-07-09 14:20:49 UTC; 27s ago
     Docs: man:nginx(8)
   Process: 9910 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 9911 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
  Main PID: 10101 (nginx)
    Tasks: 2 (limit: 4622)
   Memory: 3.0M
      CPU: 45ms
   CGroup: /system.slice/nginx.service
           |-10101 nginx: master process /usr/sbin/nginx -g daemon on; master_process on;
           `--10103 nginx: worker process

Jul 09 14:20:49 debian11 systemd[1]: Starting A high performance web server and a reverse proxy server...
Jul 09 14:20:49 debian11 systemd[1]: Started A high performance web server and a reverse proxy server.
lines 1-16/16 (END)
```

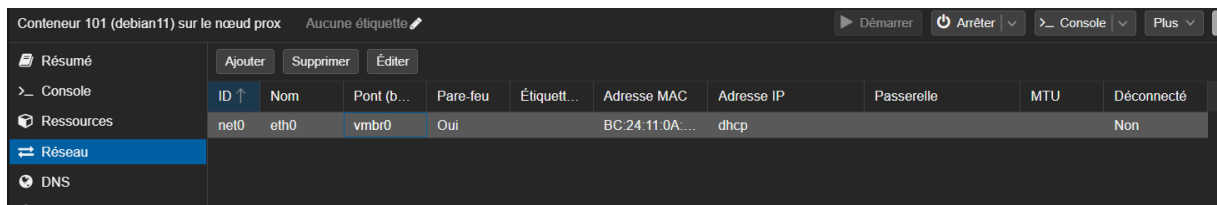
TP5 – Réseautage entre VM et conteneurs

****Configurer les VM et conteneurs sur le même bridge réseau.**

VM Ubuntu -- IP 192.168.230.136



Conteneur LXC Debian 11 – IP 192.168.230.137



**Tester la communication entre la VM et le conteneur (ping, curl sur le service Web).

PING CONTENEUR-VM :

```
root@debian11:~# ping -c 5 192.168.230.136
PING 192.168.230.136 (192.168.230.136) 56(84) bytes of data.
64 bytes from 192.168.230.136: icmp_seq=1 ttl=64 time=4.03 ms
64 bytes from 192.168.230.136: icmp_seq=2 ttl=64 time=1.43 ms
64 bytes from 192.168.230.136: icmp_seq=3 ttl=64 time=1.10 ms
64 bytes from 192.168.230.136: icmp_seq=4 ttl=64 time=1.33 ms
64 bytes from 192.168.230.136: icmp_seq=5 ttl=64 time=1.00 ms

--- 192.168.230.136 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.000/1.775/4.028/1.136 ms
root@debian11:~#
```

PING VM-CONTENEUR

```
root@ousmane-Standard-PC-i440FX-PIIX-1996:/home/ousmane# ping -c 5 192.168.230.137
PING 192.168.230.137 (192.168.230.137) 56(84) bytes of data.
64 bytes from 192.168.230.137: icmp_seq=1 ttl=64 time=0.906 ms
64 bytes from 192.168.230.137: icmp_seq=2 ttl=64 time=0.576 ms
64 bytes from 192.168.230.137: icmp_seq=3 ttl=64 time=0.589 ms
64 bytes from 192.168.230.137: icmp_seq=4 ttl=64 time=0.738 ms
64 bytes from 192.168.230.137: icmp_seq=5 ttl=64 time=0.829 ms

--- 192.168.230.137 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4059ms
rtt min/avg/max/mdev = 0.576/0.727/0.906/0.129 ms
root@ousmane-Standard-PC-i440FX-PIIX-1996:/home/ousmane#
```

CURL SUR LE SERVEUR NGINX

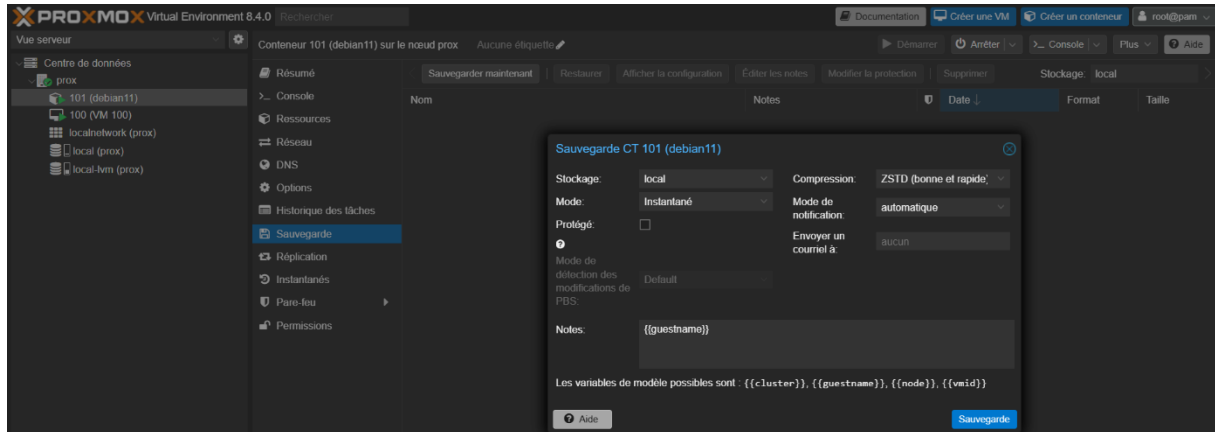
```
root@ousmane-Standard-PC-i440FX-PIIX-1996:/home/ousmane# curl http://192.168.230.137
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
    body {
        width: 35em;
        margin: 0 auto;
        font-family: Tahoma, Verdana, Arial, sans-serif;
    }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

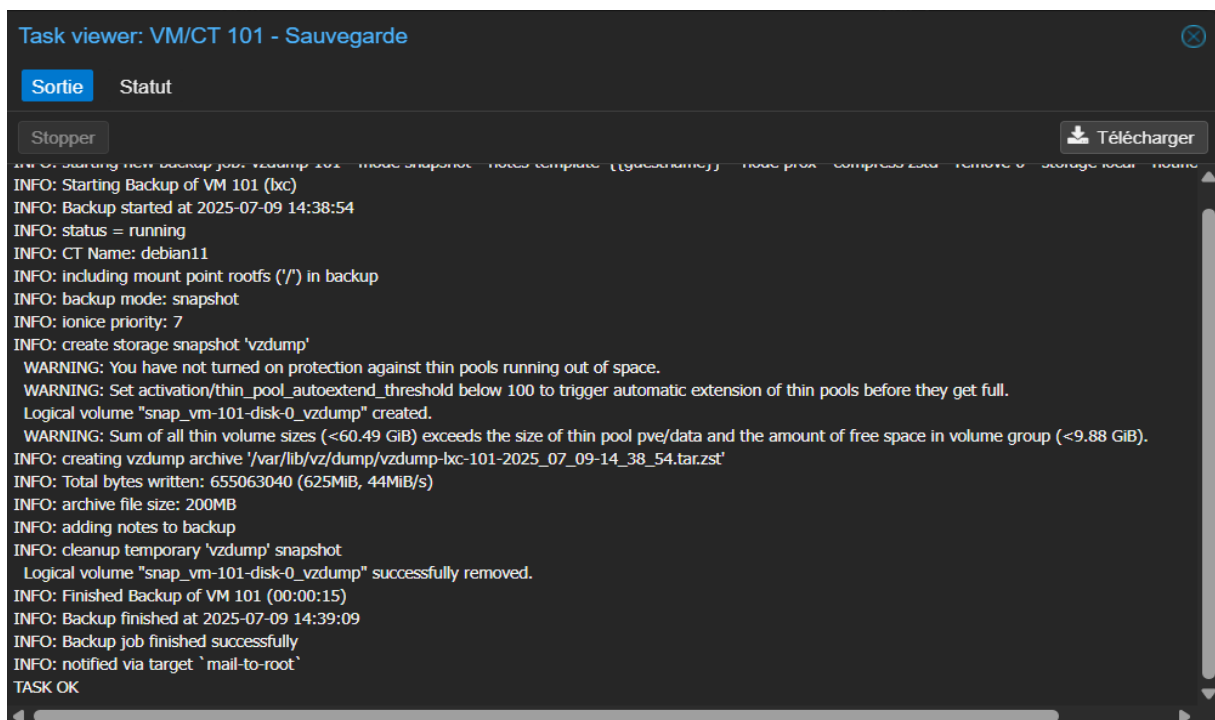
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@ousmane-Standard-PC-i440FX-PIIX-1996:/home/ousmane#
```

TP6 – Sauvegarde et restauration

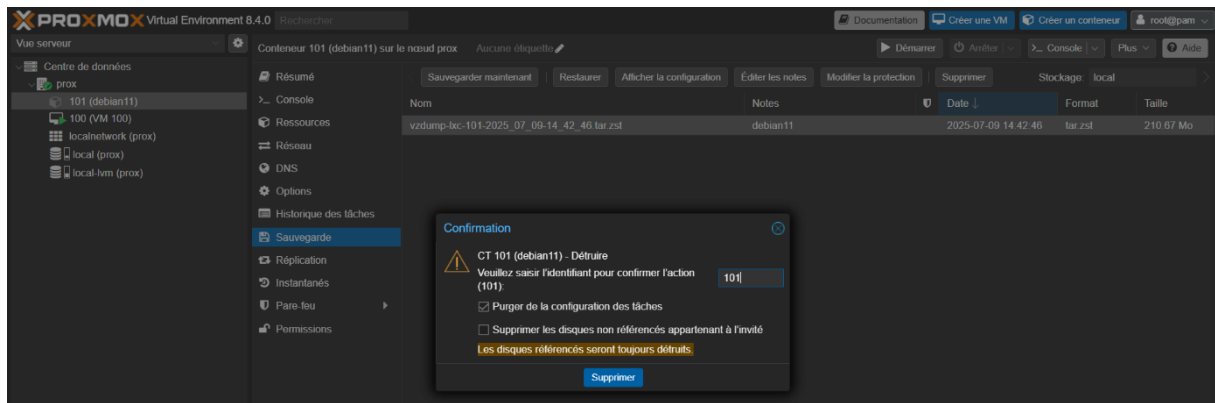
****Effectuer une sauvegarde (backup) de la VM et du conteneur.**



Le backup du conteneur a été bien accompli :



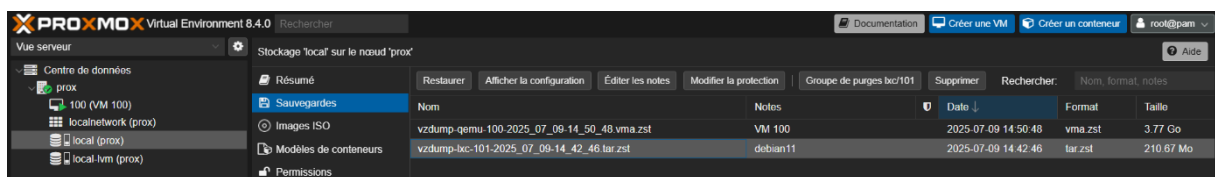
****Supprimer le conteneur et le restaurer à partir de la sauvegarde.**



On voit clairement que le conteneur 101 a été supprimé



Restauration :



Restaurer: CT

Source:

vzdump-lxc-101-2025_07_09-14_42_46.tar.zst

Stockage:

local-lvm

CT:

101

Limite de bande passante:

Par défaut, la limite de restauration du stockage cil

MiB/s

Unique:

☐

Démarrer après restauration:

☐

Niveau de privilège:

☒ Valeur configurée dans la sauvegarde

☐ Non privilégié

☐ Privilégié

Écraser les paramètres:

Nom d'hôte:

debian11

Mémoire:

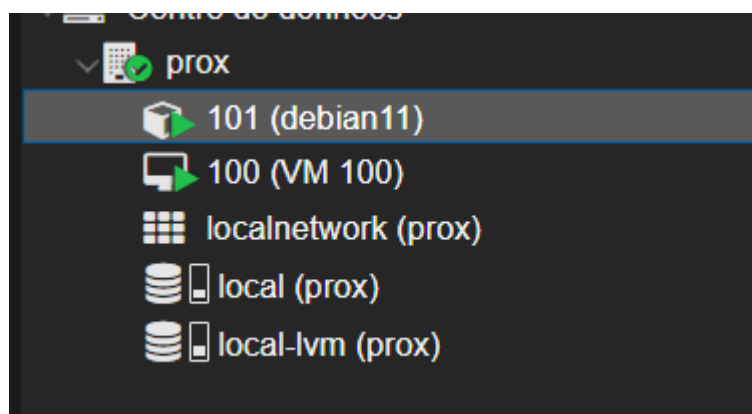
512

Cœurs:

1

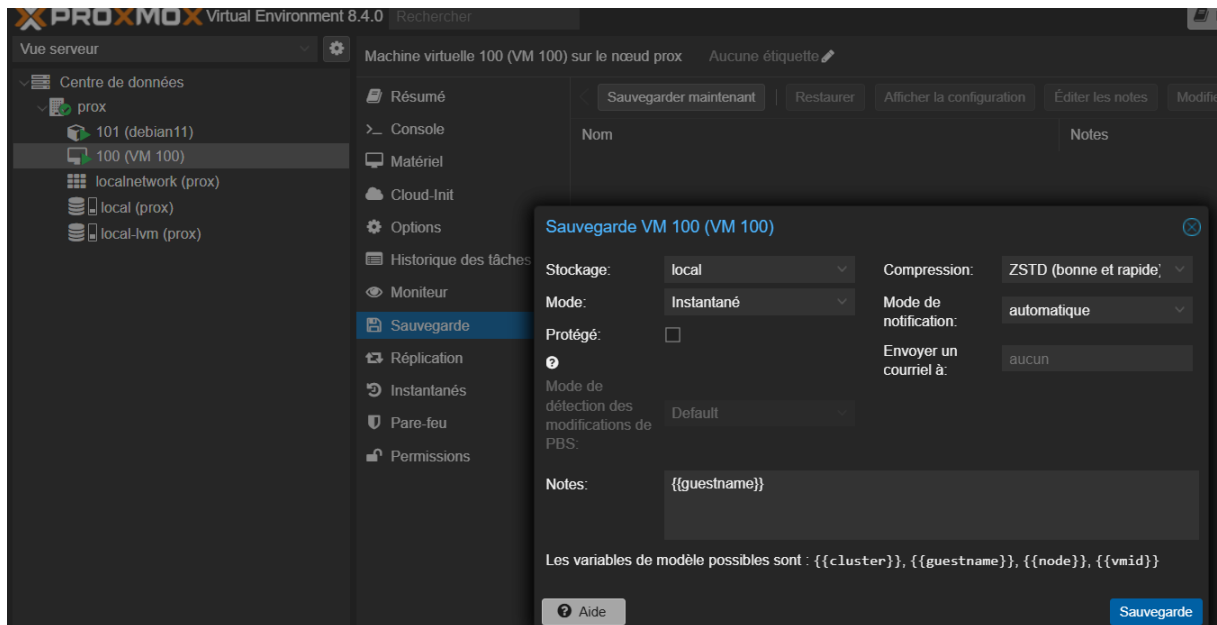
Restaurer

Le conteneur a bien été restaurer :

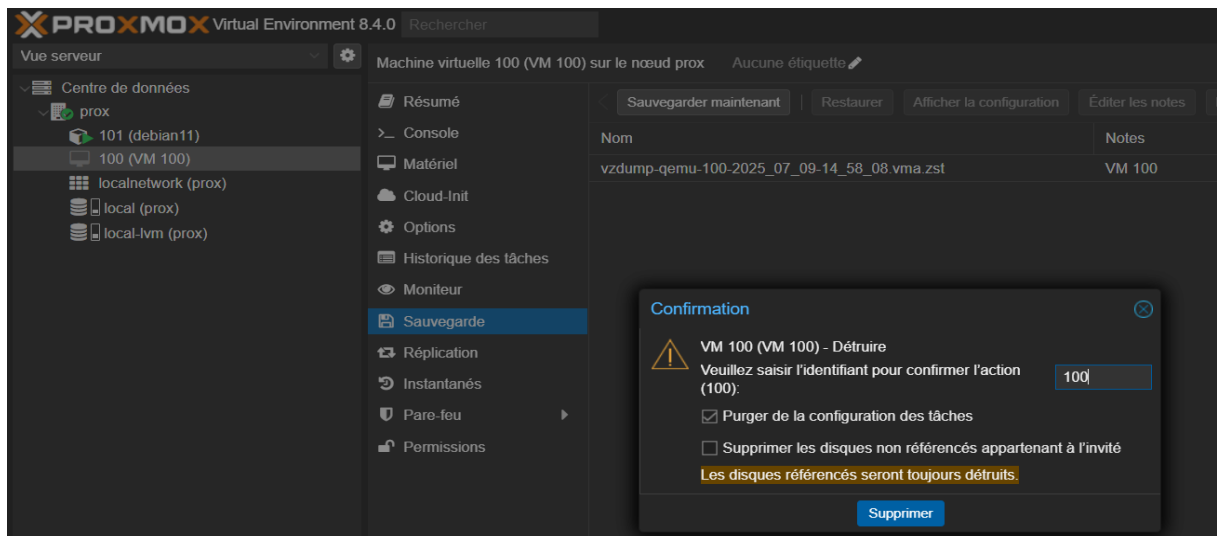


****Refaire les mêmes étapes pour la VM.**

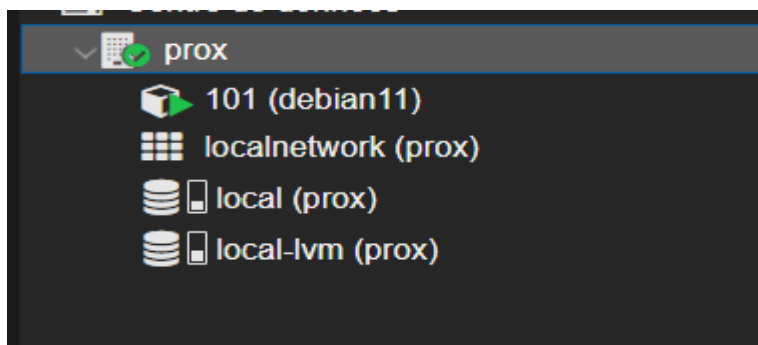
Lancement de la sauvegarde :



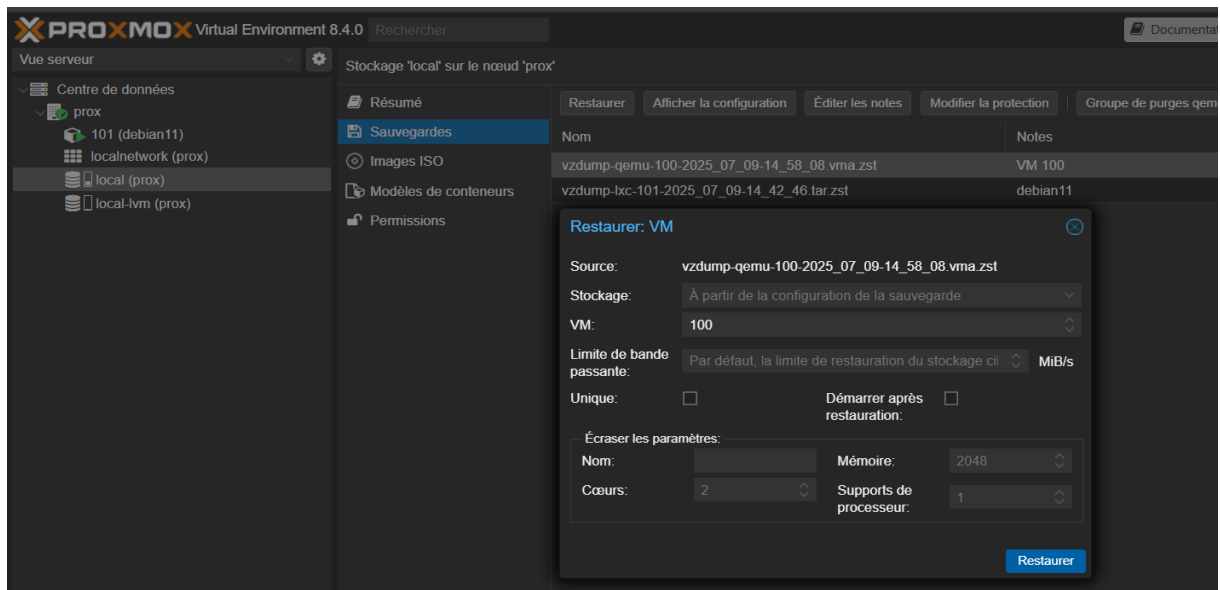
Une fois terminée on supprime la VM :



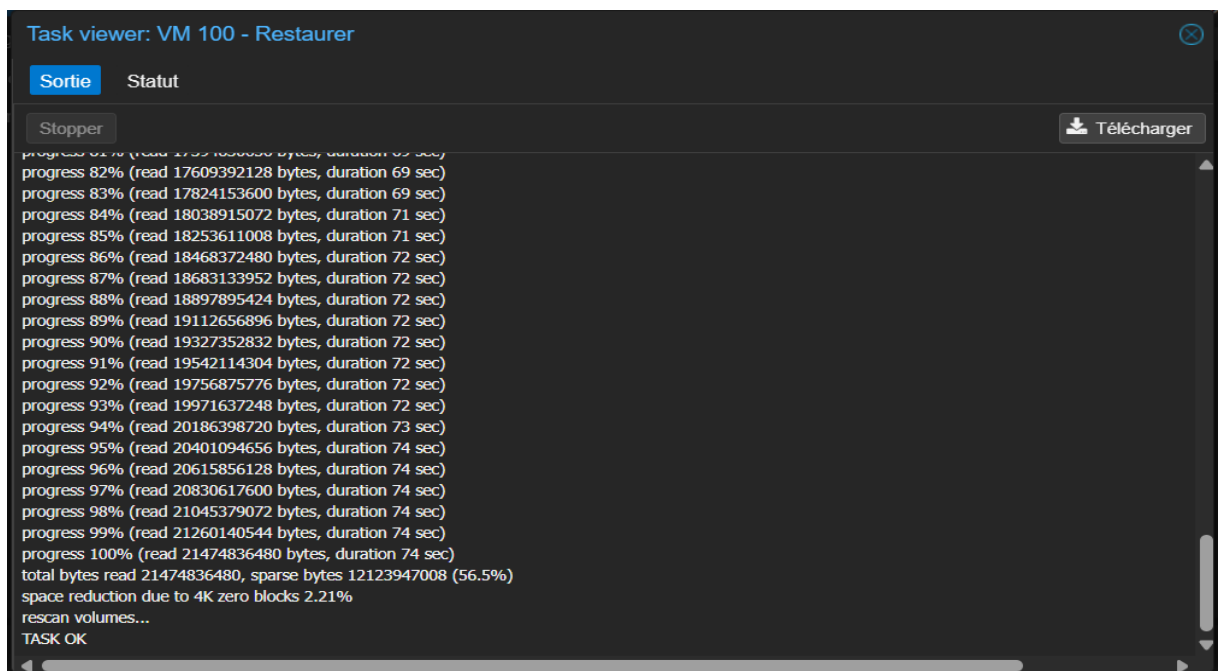
On voit clairement que le conteneur 100 a été supprimé



Ensuite on la restaure via la sauvegarde :



Puis on attend la fin de la tâche :



On voit bien que la VM a été restaurer :



TP7 – Gestion par ligne de commande (CLI)

****Créer un conteneur LXC via la CLI (commande 'pct').**

Listions d'abord les modèles disponibles

```
Administrateur : Windows PowerShell

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Jul  9 13:43:23 2025
root@prox:~# pveam available -section system
unable to open file '/var/lib/pve-manager/apl-info/releases.turnkeylinux.org' - No such file or directory
system      almalinux-9-default_20240911_amd64.tar.xz
system      alpine-3.19-default_20240207_amd64.tar.xz
system      alpine-3.20-default_20240908_amd64.tar.xz
system      alpine-3.21-default_20241217_amd64.tar.xz
system      archlinux-base_20240911-1_amd64.tar.zst
system      centos-9-stream-default_20240828_amd64.tar.xz
system      debian-11-standard_11.7-1_amd64.tar.zst
system      debian-12-standard_12.7-1_amd64.tar.zst
system      devuan-5.0-standard_5.0_amd64.tar.gz
system      fedora-40-default_20240909_amd64.tar.xz
system      fedora-41-default_20241118_amd64.tar.xz
system      gentoo-current-openrc_20231009_amd64.tar.xz
system      openeuler-24.09-default_20241118_amd64.tar.xz
system      opensuse-15.6-default_20240910_amd64.tar.xz
system      rockylinux-9-default_20240912_amd64.tar.xz
system      ubuntu-20.04-standard_20.04-1_amd64.tar.gz
system      ubuntu-22.04-standard_22.04-1_amd64.tar.zst
system      ubuntu-24.04-standard_24.04-2_amd64.tar.zst
system      ubuntu-24.10-standard_24.10-1_amd64.tar.zst
root@prox:~#
```

On va télécharger alpine3.20

```
root@prox:~# pveam download local alpine-3.20-default_20240908_amd64.tar.xz
unable to open file '/var/lib/pve-manager/apl-info/releases.turnkeylinux.org' - No such file or directory
downloading http://download.proxmox.com/images/system/alpine-3.20-default_20240908_amd64.tar.xz to /var/lib/vz/template/cache/alpine-3.20-default_20240908_amd64.tar.xz
--2025-07-09 15:22:53--  http://download.proxmox.com/images/system/alpine-3.20-default_20240908_amd64.tar.xz
Resolving download.proxmox.com (download.proxmox.com)... 185.219.221.167, 2a01:7e0:0:424::249
Connecting to download.proxmox.com (download.proxmox.com)|185.219.221.167|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3135404 (3.0M) [application/octet-stream]
Saving to: '/var/lib/vz/template/cache/alpine-3.20-default_20240908_amd64.tar.xz.tmp_dwnl.34726'
OK ..          100% 186K-16s
2025-07-09 15:23:10 (186 KB/s) - '/var/lib/vz/template/cache/alpine-3.20-default_20240908_amd64.tar.xz.tmp_dwnl.34726' saved [3135404/3135404]
calculating checksum...OK, checksum verified
download of 'http://download.proxmox.com/images/system/alpine-3.20-default_20240908_amd64.tar.xz' to '/var/lib/vz/template/cache/alpine-3.20-default_20240908_amd64.tar.xz' finished
root@prox:~#
```

Configuration du conteneur avec ses paramètres (cpu,ram,disque etc...) :

```
root@prox:~# pct create 200 \
  local:vztmpl/alpine-3.20-default_20240908_amd64.tar.xz \
  --storage local-lvm \
  --rootfs 4 \
  --memory 512 \
  --cores 1 \
  --hostname alpine-container \
  --net0 name=eth0,bridge=vbr0,ip=dhcp
Logical volume "vm-200-disk-0" created.
Creating filesystem with 1048576 4k blocks and 262144 inodes
Filesystem UUID: 31889818-aaa1-4799-ad4b-7f8337ba7a28
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736
extracting archive '/var/lib/vz/template/cache/alpine-3.20-default_20240908_amd64.tar.xz'
Total bytes read: 10864640 (11MiB, 19MiB/s)
Detected container architecture: amd64
root@prox:~#
```

****Lancer, arrêter, supprimer un conteneur via CLI.**

1. Lancement du conteneur :

```
root@prox:~# pct start 200
root@prox:~# pct list
VMID      Status    Lock      Name
101       running             debian11
200       running             alpine-container
root@prox:~#
```

Etat :



2. Arrêt du conteneur :

```
root@prox:~# pct stop 200
root@prox:~# pct list
VMID      Status    Lock      Name
101       running             debian11
200       stopped             alpine-container
root@prox:~#
```

Etat :



3. Suppression du conteneur :

```
root@prox:~# pct destroy 200
Logical volume "vm-200-disk-0" successfully removed.
root@prox:~# pct list
VMID      Status    Lock      Name
101       running             debian11
root@prox:~#
```

Etat :



**Lister les ressources (VM, CT, stockage) avec `qm`, `pct`, et `pvesh`.

- Lister les VMs :

```
root@prox:~# qm list
      VMID NAME                STATUS    MEM(MB)    BOOTDISK(GB) PID
      100 VM 100                running    2048         20.00 30843
root@prox:~#
```

- Lister les conteneurs :

```
root@prox:~# pct list
VMID      Status    Lock      Name
101       running
root@prox:~#
```

- Lister les stockages :

```
root@prox:~# pvesh get /storage
storage
local
local-lvm
root@prox:~#
```

- Lister les noeuds :

```
root@prox:~# pvesh get /nodes
```

node	status	cpu	level	maxcpu	maxmem	mem	ssl_fingerprint	uptime
prox	online	1.23%		4	3.83 GiB	3.03 GiB	52:62:5D:B2:FC:FE:CF:17:AB:C2:80:43:80:F8:33:C2:78:14:79:5E:3F:27:A7:4B:E0:15:12:49:44:88:6D:92	1h 49m 54s

```
root@prox:~#
```